Global reciprocity on principal idèles and the sum of local invariants

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Source: *Class Field Theory*, v4.03, 6 August 2020

Private reading transcription of the author's Theorem VII.8.1, its supplied cyclotomic proof, and the necessary Brauer-group support in VII§7. The selected core occupies physical PDF pages 228–232 (printed pages 219–223). The local cyclotomic formulas used in Example VII.8.2 are retained from I, Example 3.13, physical page 53 (printed page 44), with the separate published correction identified below. The complete fixed original is retained at https://www.jmilne.org/math/CourseNotes/CFT.pdf. Mathematical expressions and the author's argument are retained, with connective prose and layout normalized. This is published-PDF transcription, not author-native TeX or a new proof.

Notation and preceding source results

For a number field K, $\mathbb{I}_K$ is its idèle group and $\mathbf{C}_K = \mathbb{I}_K/K^\times$. Write

$$H^2(L/K) = H^2(\mathrm{Gal}(L/K), L^\times), \qquad H^2(/K) = H^2(\mathrm{Gal}(K^{\mathrm{al}}/K), (K^{\mathrm{al}})^\times).$$

As explained in Chapter IV, these identify with $\mathrm{Br}(L/K)$ and $\mathrm{Br}(K)$. For a place v of K, L^v denotes the completion at a place above v; G^v is the decomposition group. The local image of a cohomology class α is denoted α_v. The source abbreviates $\mathrm{inv}_v(\alpha_v)$ to $\mathrm{inv}_v(\alpha)$. The final sentence of VII.8.6 writes $H^2(K)$ after earlier using $H^2(/K)$ for its ambient absolute class; that printed notation is retained.

For a finite abelian extension L/K with Galois group G, the author has defined the homomorphism

$$\phi_{L/K} : \mathbb{I}_K \longrightarrow G, \qquad \phi_{L/K}(\mathbf{a}) = \prod_v \phi_v(a_v) \quad (\mathbf{a} = (a_v)),$$

from the local Artin maps. The background used here includes local class field theory and its invariant map, the decomposition of idèle cohomology (VII.2.5), cup products, Hilbert 90, and inflation–restriction. The preceding second inequality (VII.5.1) gives $H^1(G, \mathbf{C}_L) = 0$ and

$$(\mathbb{I}_K : K^\times \mathrm{Nm}_{L/K}\, \mathbb{I}_L) \mid [L : K].$$

Its complete algebraic proof is a separate, linked reader in this same source work; that core is not counted again here. In the proof of VII.8.5 the author writes the abelian target group additively; there $\phi_{L/K}(a) = 0$ denotes its identity element.

The target statement

Theorem VII.8.1. *(a) Let L/K be a finite abelian extension of number fields. Then $\phi_{L/K}$ takes the value 1 on the principal idèles $K^\times \subset \mathbb{I}_K$.*

(b) Let L/K be a finite Galois extension of number fields. Then

$$\sum_v \mathrm{inv}_v(\alpha) = 0 \qquad \textit{for every } \alpha \in H^2(L/K).$$

The author's Brauer-group support

Theorem VII.7.1. *For every Galois extension L/K of number fields, possibly infinite, the canonical map*

$$H^2(L/K) \longrightarrow \bigoplus_v H^2(L^v/K_v)$$

is injective.

Proof. First suppose that L/K is a finite Galois extension with Galois group G. Because $H^1(G, \mathbf{C}_L) = 0$, the cohomology sequence of

$$0 \longrightarrow L^\times \longrightarrow \mathbb{I}_L \longrightarrow \mathbf{C}_L \longrightarrow 0$$

gives

$$0 \longrightarrow H^2(G, L^\times) \longrightarrow H^2(G, \mathbb{I}_L) \longrightarrow \cdots .$$

But $H^2(G, L^\times) = H^2(L/K)$ and, by VII.2.5,

$$H^2(G, \mathbb{I}_L) = \bigoplus_v H^2(G^v, L^{v\times}) = \bigoplus_v H^2(L^v/K_v),$$

which proves the finite case. For an infinite extension, pass to the limit over its finite Galois subextensions. □

An extension L/K is called *cyclotomic* if $L \subset K[\zeta]$ for some root of 1, ζ.

Proposition VII.7.2. *For every $\beta \in H^2(/K)$, there is a cyclic cyclotomic extension L of K such that β maps to zero in $H^2(/L)$.*

Proof. Theorem VII.7.1 shows that β is determined by its images in $H^2(K_v)$, and hence by $\mathrm{inv}_v(\beta_v) \in \mathbb{Q}/\mathbb{Z}$ (III, Theorem 2.1). For every finite extension L/K and prime w of L above v,

$$\mathrm{inv}_w(\beta|_L) = [L_w : K_v]\,\mathrm{inv}_v(\beta).$$

Thus we must find a cyclic cyclotomic L/K such that

$$[L^v : K_v]\,\mathrm{inv}_v(\beta_v) = 0 \pmod{\mathbb{Z}} \quad \text{for all } v.$$

Because the map from $H^2(L/K)$ into $H^2(/K)$ maps into the direct sum of the local groups, $\mathrm{inv}_v(\beta_v) = 0$ for almost all v. Consequently some integer $m > 0$ satisfies $m\,\mathrm{inv}_v(\beta_v) = 0$ for every v. The next lemma supplies L. □

Lemma VII.7.3. *Given a number field K, a finite set S of finite primes of K, and an integer $m > 0$, there is a totally complex cyclic cyclotomic extension L of K such that $m \mid [L^v : K_v]$ for every $v \in S$.*

Proof. It suffices to prove the assertion for $\mathbb{Q}$ and $m[K : \mathbb{Q}]$, so assume $K = \mathbb{Q}$. Let ℓ be prime and let ζ be a primitive ℓ^rth root of 1 with $r > 2$. Then

$$\mathrm{Gal}(\mathbb{Q}[\zeta]/\mathbb{Q}) \simeq (\mathbb{Z}/\ell^r\mathbb{Z})^\times,$$

and

$$(\mathbb{Z}/\ell^r\mathbb{Z})^\times \simeq \begin{cases} \Delta \times C(\ell^{r-1}) & \ell \text{ odd}, \\ \Delta \times C(2^{r-2}) & \ell = 2. \end{cases}$$

Here Δ has order $\ell-1$ and 2 in the two cases, and $C(t)$ is a cyclic group of order t (Serre 1970, II, 3.2). Therefore $L(\ell^r) \stackrel{\text{def}}{=} \mathbb{Q}[\zeta]^\Delta$ is a cyclic cyclotomic extension of $\mathbb{Q}$ of degree ℓ^{r-2} or ℓ^{r-3}. [1]

Consider $\mathbb{Q}_p[\zeta]$. If $p=\ell$, $\mathbb{Q}[\zeta]$ is totally ramified over p, and

$$[\mathbb{Q}_p[\zeta]:\mathbb{Q}_p] = [\mathbb{Q}[\zeta]:\mathbb{Q}] = \varphi(\ell^r).$$

If $p\neq\ell$, $\mathbb{Q}[\zeta]$ is totally unramified over p, and $[\mathbb{Q}_p[\zeta]:\mathbb{Q}_p]=t$, where t is the smallest integer such that $\ell^r \mid p^t-1$. In either case, this local degree tends to infinity as $r\to\infty$. Thus for every p, $[L(\ell^r)^p:\mathbb{Q}_p]$ is a power of ℓ tending to infinity with r.

A product of cyclic groups of distinct prime-power orders is cyclic. Therefore, for distinct primes $\ell_1,\dots,\ell_s$,

$$L = L(\ell_1^{r_1})\cdots L(\ell_s^{r_s})$$

is cyclic. By choosing $\ell_1^{r_1}\cdots\ell_s^{r_s}$ sufficiently large, the local degrees $[L^p:\mathbb{Q}_p]$ are divisible by m for all $p\in S$. □

The author expresses these results more concretely: a central simple algebra over K that splits over every K_v splits over K; every central simple algebra over K splits over a cyclic cyclotomic extension of K.

The local formulas used by the author

The following are the portions of I, Example 3.13 used in VII, Example 8.2. They describe the local Artin map $\phi_p:\mathbb{Q}_p^\times\to\mathrm{Gal}(L/\mathbb{Q}_p)$ when $L=\mathbb{Q}_p[\zeta]$ and ζ is a primitive nth root of 1.

I.3.13(a). If n is prime to p, then $L/\mathbb{Q}_p$ is unramified, of degree equal to the residue-field degree. The residue field is $\mathbb{F}_{p^f}$, where p^f is the smallest power of p with $n\mid p^f-1$. The map sends up^t to the tth power of Frobenius, and its kernel is $\mathbb{Z}_p^\times\langle p^f\rangle$.

I.3.13(b), the used formula as printed. When $n=p^r$, write $a=up^t$, and let $u_0\in\mathbb{Z}$ represent the class of u in $(\mathbb{Z}_p/p^r\mathbb{Z}_p)^\times$. The original page writes that $\phi_p(a)$ sends ζ to $\zeta^{u^{-1}}$, and gives its kernel as

$$\{up^m \mid u\equiv 1 \pmod{p^r},\ m\in\mathbb{Z}\}.$$

Only the local-action formulas required for the selected proof are quoted here; the complete example remains in the original fixed work.

Separate published correction. Under the CFT v4.03 section of the author's errata page, the typographical note attributed to Raghuram Sundararajan says that the last line of I.3.13(b) should use

$$\phi_p(a):\zeta\longmapsto\zeta^{u_0^{-1}}.$$

The original and the correction are kept distinct. The complete saved page is https://www.jmilne.org/math/CourseNotes/errata.html. The correction to composite part (c) remains in that saved source; this reader uses the prime-power route and does not transcribe (c).

[1] Source wording note: the degrees ℓ^{r-2} and ℓ^{r-3} are the exponents printed on physical page 229 (printed page 220). This transcription retains that source sentence and supplies no replacement degree formula or claimed author erratum.

The author's proof: the cyclotomic case

Example VII.8.2. *We verify part (a) for $\mathbb{Q}[\zeta_m]/\mathbb{Q}$, where ζ_m is a primitive mth root of 1.*

Identify $\mathrm{Gal}(\mathbb{Q}[\zeta_m]/\mathbb{Q})$ with $(\mathbb{Z}/m\mathbb{Z})^\times$. Here $a \in \mathbb{Q}^\times$ is viewed as a principal idèle, as in part (a). For n relatively prime to m, $[n]$ denotes the automorphism sending ζ_m to ζ_m^n. It suffices to show $\phi(a)|_{\mathbb{Q}[\zeta_{\ell^r}]} = 1$ for all $\ell \mid m$. We may assume $m = \ell^r$, $m \neq 2$.

The homomorphism

$$\phi_\infty : \mathbb{R}^\times / \mathrm{Nm}(\mathbb{C}^\times) \longrightarrow \mathrm{Gal}(\mathbb{Q}[\zeta_m]/\mathbb{Q})$$

sends a negative real number to complex conjugation, so $\phi_\infty(a) = [\mathrm{sign}(a)]$. Write $a = up^s \in \mathbb{Q}_p^\times$. If $p \neq \ell$, then p is unramified in $\mathbb{Q}[\zeta_m]$ and $\phi_p(a)$ acts as the sth power of Frobenius at p:

$$\phi_p(up^s) = [p^s].$$

The prime ℓ is totally ramified in $\mathbb{Q}[\zeta_m]$, and

$$\phi_\ell(a) = [u^{-1}]$$

by I.3.13. [2]

It suffices to show $\phi(a) = 1$ in the three cases $a = -1$, $a = \ell$, and $a = q$ for a prime $q \neq \ell$. The author gives

$$\phi_p(-1) = \begin{cases} [-1] & p = \infty, \\ [-1] & p = \ell, \\ [1] & p \neq \ell, \infty, \end{cases} \qquad \phi_p(\ell) = \begin{cases} [1] & p = \ell, \\ [1] & p \neq \ell, \end{cases}$$

and

$$\phi_p(q) = \begin{cases} [q] & p = q, \\ [q^{-1}] & p = \ell, \\ [1] & p \neq \ell, q. \end{cases}$$

In every case, $\prod_p \phi_p(a) = 1$.

Lemma VII.8.4.

(a) If part (a) holds for L/K, it holds for every subextension.

(b) If part (a) holds for L/K, it holds for LK'/K' for every number field $K' \supset K$.

Proof. (a) Suppose $L \supset K' \supset K$. Then $\phi_{K'/K}$ is the composite of $\phi_{L/K}$ and restriction $\mathrm{Gal}(L/K) \to \mathrm{Gal}(K'/K)$, because this is true for the local Artin maps.

(b) Put $L' = LK'$. For each prime w of K', there is a commutative diagram

$$\begin{array}{ccc} {K'_w}^\times & \xrightarrow{\phi_w} & \mathrm{Gal}(L'^w/K'_w) \\ \Big\downarrow{\scriptstyle \mathrm{Nm}} & & \Big\downarrow \\ K_v^\times & \xrightarrow{\phi_v} & \mathrm{Gal}(L^v/K_v). \end{array}$$

[2] The source records Catherine O'Neil's observation that this uses I§3 despite the earlier promise that I§§2–4 could be skipped. The author mentions an elementary alternative and leaves its clarification to a future version. The supplied local-Artin route is retained here.

Combining these gives the commutative diagram

$$\begin{array}{ccc} \mathbb{I}_{K'} & \xrightarrow{\phi_{L'/K'}} & \mathrm{Gal}(L'/K') \\ {\scriptstyle \mathrm{Nm}}\downarrow & & \downarrow \\ \mathbb{I}_K & \xrightarrow{\phi_{L/K}} & \mathrm{Gal}(L/K). \end{array}$$

The norm map on idèles sends $K'^{\times}$ into $K^{\times}$, so the lemma follows from the earlier results. □

Example VII.8.2 and Lemma VII.8.4 show that part (a) holds for all cyclotomic extensions of a number field K.

The relation between the two statements

Lemma VII.8.5. *Let L/K be an abelian extension of number fields. If part (b) holds for L/K, then part (a) also holds. Conversely, if L/K is cyclic and part (a) holds, then part (b) also holds.*

Proof. Let $\chi \in \mathrm{Hom}(G, \mathbb{Q}/\mathbb{Z})$. Regard χ as an element of $H^1(G, \mathbb{Q}/\mathbb{Z})$. Its image under the boundary map from

$$0 \longrightarrow \mathbb{Z} \longrightarrow \mathbb{Q} \longrightarrow \mathbb{Q}/\mathbb{Z} \longrightarrow 0$$

is $\delta\chi \in H^2(G, \mathbb{Z})$. Consider

$$\begin{array}{ccccc} K^{\times} & \longrightarrow & \mathbb{I}_K & \xrightarrow{\phi_{L/K}} & G \\ {\scriptstyle \cup\delta\chi}\downarrow & & \downarrow{\scriptstyle \cup\delta\chi} & & \downarrow{\scriptstyle \chi} \\ H^2(G, L^{\times}) & \longrightarrow & H^2(G, \mathbb{I}_L) & \longrightarrow & \mathbb{Q}/\mathbb{Z}. \end{array}$$

The first two vertical arrows are cup product by $\delta\chi$. If $\delta\chi$ is represented by the 2-cocycle $n_{\sigma,\tau}$, then the image of x is represented by $\sigma, \tau \longmapsto x^{n_{\sigma,\tau}}$. The left square therefore commutes. The right vertical arrow is χ itself; commutativity of the right square follows from III.3.6.

Assume part (b) for L/K. Then $\chi(\phi_{L/K}(a)) = 0$ for all characters χ of G, so $\phi_{L/K}(a) = 0$. Conversely, when G is cyclic choose χ injective. Then (a) implies (b), since $\cup\delta\chi$ is an isomorphism (II, 3.5). □

Part (a) is known for cyclotomic extensions, hence part (b) is known for cyclic cyclotomic extensions. Proving part (b) proves the whole theorem. The next result therefore completes its proof.

Lemma VII.8.6. *If part (b) holds for cyclic cyclotomic extensions, then it holds for all finite Galois extensions.*

Proof. Let $\beta \in H^2(/K)$. We are given that if $\beta \in H^2(L/K)$ for some cyclic cyclotomic L/K, then $\sum_v \mathrm{inv}_v(\beta) = 0$, where β_v is the image of β in $H^2(K_v)$. Proposition VII.7.2 says that every $\beta \in H^2(K)$ lies in $H^2(L/K)$ for such a cyclotomic extension.

The source appends the following bracketed explanation:

> We use the inflation–restriction sequence
>
> $$0 \longrightarrow H^2(\mathrm{Gal}(L/K), L^{\times}) \longrightarrow H^2(G_K, \overline{K}^{\times}) \longrightarrow H^2(G_L, \overline{L}^{\times}),$$
>
> and β is in the middle group and vanishes in the right-hand group.

□

The author's reciprocity quotient consequence

The author explains why part (a) gives the Reciprocity Law for the finite abelian extension L/K in that part. It says that $K^\times$ is in the kernel of $\phi_{L/K} : \mathbb{I}_K \to \mathrm{Gal}(L/K)$. The kernel already contains $\mathrm{Nm}_{L/K}\,\mathbb{I}_L$, because this holds locally. Thus there is a homomorphism

$$\mathbb{I}_K/(K^\times \mathrm{Nm}_{L/K}\,\mathbb{I}_L) \longrightarrow \mathrm{Gal}(L/K).$$

For a finite prime v unramified in L, an idèle with a prime element in the v-position maps to its Frobenius element. Proposition VII.4.7 therefore gives surjectivity. The second inequality (VII.5.1) gives

$$(\mathbb{I}_K : K^\times \mathrm{Nm}_{L/K}\,\mathbb{I}_L) \leq [L : K],$$

so this homomorphism is an isomorphism.

The single collected target is VII.8.1 with its two differently scoped parts. The Brauer injectivity, cyclic cyclotomic splitting, local-action context, cup-product bridge and quotient consequence are support for that one target. VII.5.1 is linked prior support, not an additional count here. The alternative route mentioned in Remark VII.8.3 is not added as a second proof. This reader does not enlarge the target to the full Brauer–Hasse–Noether exact sequence.